

PHYS 341 - Quantum Lab II

Lab 1: Pulsed Nuclear Magnetic Resonance

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1 Abstract

Pulsed Nuclear Magnetic Resonance (PNMR) is an analysis method which can be used to study the internal structure of materials. This report describes the background of the subject, and the experimental process and results of measuring the free induction decay (FID / T_2^*), spin-lattice relaxation time, T_2 , and the spin-spin relaxation time, T_1 , in a sample of mineral oil. This was accomplished by utilizing features of nuclear magnetic resonance such as the spin echoes and flips.

2 Background

The interactions of atoms both with each other and with internal particles are subjects of keen interest in physics as they provide insight into the building blocks of matter and the universe. Over time scientists have developed techniques for imaging and observing these interactions using electromagnetic and nuclear physics. The existence of a nuclear magnetic moment has allowed physicists to manipulate atoms in ways that can be detected electrically, and these developments have led to the creation of scientific life-saving medical advancements such as the MRI machine, which uses NMR imaging to produce a detailed non-invasive 3-dimensional picture of a patient's inner workings, allowing physicians to better diagnose and treat medical issues.

2.1 Nuclear Magnetic Resonance

Nuclear Magnetic Resonance (NMR) is a technique which employs external magnetic fields and radio-frequency pulses to align the spins of protons in atoms' nuclei. Protons have a spin, which is to say an angular momentum, and since they are electrically charged, a magnetic moment as well [1]. The direction of this spin and magnetic moment can be altered using NMR.

A proton's magnetic moment μ can be related to its spin J :

$$\mu = \gamma J \quad (1)$$

where γ is the gyromagnetic ratio. For protons, this is calculated to be $\gamma_{p+} = 2.675 * 10^4 \text{ rad sec}^{-1} \text{ gauss}^{-1}$.

When placed in an external magnetic field, the proton will experience a torque τ , which causes a rotation in the direction of the magnetic moment:

$$\vec{\tau} = \vec{\mu} \times \vec{B} = \frac{dJ}{dt} \quad (2)$$

When relating (2) to (1), we can see that

$$\vec{\mu} \times \vec{B} = \frac{1}{\gamma} \frac{d\mu}{dt} \quad (3)$$

If $\vec{\mu}$ is aligned with $\vec{B}$, then there is no torque. But, if there is a difference θ between the two, it experiences a torque. This torque causes a precession of the direction of the proton's magnetic moment. This precession is char-

acterized by the rotation of the magnetic moment around the axis of the direction of the magnetic field, as seen in Figure 1 below.

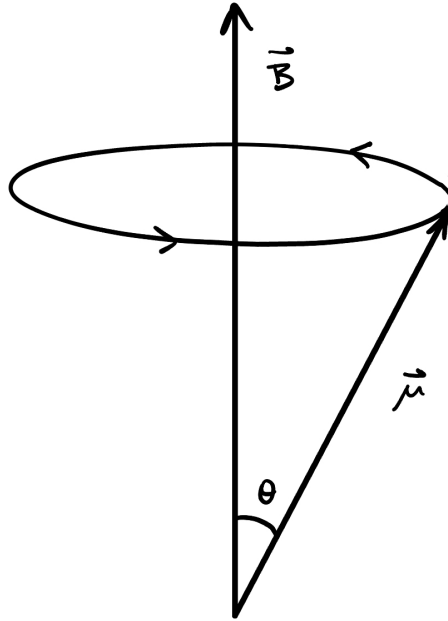


Figure 1: Magnetic moment μ of particle precessing around magnetic field B

This precession is analogous to classical spin, where a spinning object with mass, such as a gyroscope, whose angular momentum is misaligned with the gravitational field will precess around the axis of that field.

The direction of precession is specified by the right hand rule (RHR), which shows a particle in a magnetic field pointing to the top of the page with magnetic moment pointing to the left of the page will experience a torque to the normal of the page, going out towards the reader. If the particle is glued to the page, once the moment is pushed a little, there will be a new

axis between μ and B , meaning it experiences a torque to the normal of that new axis. And so on and so on, this process repeats, causing the rotation around the axis.

Understanding precession is vital to accomplishing this experiment, as it is the main feature utilized in Pulsed Nuclear Magnetic Resonance, which will allow us to study characteristics of atoms difficult to examine otherwise.

2.2 Pulsed NMR and Relaxation

On its own, nuclear magnetic resonance is a powerful tool for observing the atoms and molecular structure. However, to measure interactions between these tiny particles requires observation of the atoms without the influence of an oscillating magnetic field [2]. Once the strong external field has been applied, and the sample has been magnetized along that axis, the oscillating field that creates resonance can be applied in pulses, rather than continuously. After each pulse, internal interactions cause the nuclear magnetic moments to 'relax' back to lower energy states, something which we can observe electrically.

The two interactions of interest are the **spin-spin relaxation time**, denoted T_2 , and the **spin-lattice relaxation time**, denoted T_1 . These are properties of the interactions between atoms, nuclei, and smaller particles, and provide information on the relative strengths of the interactions. However, due to the inhomogeneous nature of the static magnetic field, these times are difficult to observe.

In PNMR experiments a large static magnetic field is applied to the sample, which we will call B_0 , and we say it is oriented in the z -direction. When an oscillating magnetic field is applied (with much less strength) to the sample, nuclear spins begin to transition to higher energy states and rotate down to the xy plane. The electrical signal we can measure can be interpreted as the collective sum of the xy components of all the individual nuclear magnetic moments. Now once the total magnetic moment $\vec{M}$ has an xy component, it begins precessing around the z -axis, due to the way magnetic torque and angular momentum work classically [3]. If one looks deeper, they would see that each individual nuclear magnetic moment is precessing around the z axis, since that is the orientation of external static field. The speed of this precession is related to the strength of the field, and this is where inhomogeneity comes into effect. Since the field has slightly different strengths at different positions, some nuclear spins precess at different speeds than others. At some point, the orientations of all the nuclear spins have spread and cancel each other out, such that the xy component of the total magnetic moment is zero, and no electrical signal is measured.

The problem arises because the time it takes for the spins to spread and cancel out the electrical signal is much faster than the relaxation times we wish to measure. We call this time the $T2^*$ decay time, and it dominates the observed electrical signal. The challenge is now to isolate the relaxation times in order to see the desired interactions.

2.2.1 T2 and the Spin Echo

The spin-spin relaxation time is a property of the interactions of nuclear magnetic spins with nearby electrons. The existence of a fluctuating electron spin nearby a nuclear spin that has been rotated into the xy plane will cause the nuclear spin to 'decay' back to the z -axis. As stated above, the electric signal shows the xy -component of the total magnetic moment over time, but is dominated by the T_2^* decay so we cannot see how fast the magnetic moments decay back to the z axis.

T_2 is also called the *dephasing time* sometimes, and we can measure it by generating something called a **spin echo**. Suppose our external field is aligned in the z direction. When the sample is placed in this field, the nuclear spins will tend to align in that direction, creating a total magnetic moment $\vec{M}$ in the z direction. If an NMR pulse is applied for just the right amount of time, the total magnetic moment can be rotated into the xy plane. This is called a 90 degree pulse. As soon as precession begins, the spins will spread out, causing the xy signal to decay to zero in the T_2^* effect. If one then applies a 180 degree pulse, flipping the spins, then the directions of precession will be reversed. The fast precessing spins will be precessing towards the slower ones, and the spins will align briefly in a complete reversal of the spreading out from before. This alignment will cause a peak called a **spin echo**. The amplitude of this peak is governed by the dephasing process and therefore allows us to isolate T_2 . Delaying the second pulse further will change the height of the echo, and the relationship between delay time and

echo amplitude will provide an exponential relationship with T2.

2.2.2 T1 and Thermodynamics

The slowest way that the magnetic moments decay back to the z -axis is due to thermodynamics. When the sample is placed in the external static field, it becomes magnetized in the z -direction. At this point, the sample is at thermodynamic equilibrium. The external field creates two states, up and down spin, for the particles, and in accordance with the Boltzmann Distribution, one can calculate the equilibrium magnetization to be aligned with the external field. If the orientation of $\vec{M}$ is disturbed, with pulsed NMR, then it slowly returns to this thermodynamic equilibrium exponentially with a time constant we call T1. The relaxation time in this situation is slower than both T2* and T2 and is therefore dominated by both. The technique to observe this parameter is to take advantage of the rotation of $\vec{M}$. We know the relationship between M_z and time, but we can only measure M_{xy} .

Suppose $\vec{M}$ is oriented in the z direction, and we apply a 180 degree pulse. Then $\vec{M}$ will orient in the negative z direction, and begin thermodynamically decaying towards the positive z direction. After a delay time, we can measure the M_z component by applying a 90 degree pulse. Then the M_z component will be in the xy -plane and it will register an electrical signal. Measuring the height of this signal at different times will allow us to experimentally find M_z at different times after the 180 degree pulse.

2.3 Experimental Parameters

To perform the experiment and gather the necessary data certain values are needed to tune the input to the correct values. Firstly, we need to perform a 90 degree pulse in order to obtain the most accurate data. To find the correct pulse width we perform the calculation:

$$t_w = \frac{\pi}{2\gamma B_1}$$

For a static magnetic field $B_1 = 3600\text{ G}$ the pulse width is calculated to be $t_w = 4.9\mu s$. We could also see this was the correct width as the peak of the FID signal was at a maximum.

In addition, we'd like to have the internal frequency of the rotating magnetic fields be the resonant frequency of the precessing magnetic moments. This frequency was experimentally determined to be around 15.4 MHz. The experimental setup included a mixer which would output a signal demonstrating the difference between those two frequencies. The output frequency was adjusted so that the mixer output went to zero, with no beats in the waveform.

3 Technical Questions

3.1 T2*: Magnetic Field Inhomogeneity

In order to measure the FID constant, or T2*, a single A pulse was sent. The frequency of the pulse was adjusted until the Voltage of the signal displayed on the oscilloscope was seen going to 0 as quickly as possible, as seen by feature (2) on the blue signal in Figure (2) below. This rapid decay was sought to ensure the signal was at resonance, meaning it would display the maximum amplitude of the signal prior to its decay, allowing for the most clear measurements.



Figure 2: Oscilloscope display, showing A pulse and its effects on the signal

The green and blue signals, channels 2 and 3 on the oscilloscope, are both

representations of the same signal. The green signal is the 'envelope' of the blue signal, which is to say it is a representation of the maximum amplitude of the signal. It shows the decay in a much more clear line, rather than have the signal jump up and down, as seen in Figure (3) below, where the frequency was put out of phase and the voltage ranges adjusted to better show the relationship between the channels.

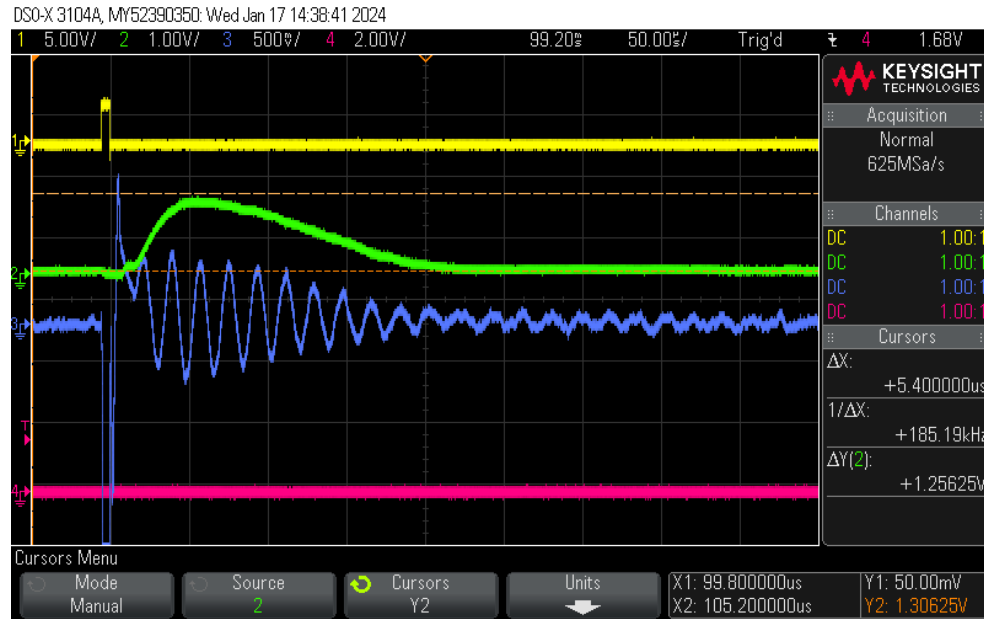


Figure 3: Oscilloscope display, showing the relationship between signal (blue) and it's envelope (green)

The yellow line at the top shows the A pulse sent out by the pulse programmer. The A pulse seen in Figure (2) rotates the proton moments 90° 's. This rotation is not instantaneous, and is seen in the sudden increase in the envelope's slope seen at feature (1) on the figure. At it's peak, the sample's protons' moments are aligned in the same direction in the xy plane, with no

z component anymore. As soon as they reach the bottom of the rotation, the external magnetic field induced by the A pulse 'lets them go', and they begin precessing. Due to the impurities in the metal of the permanent magnet, the magnetic field is inhomogeneous, with different parts of the field having different strengths. Due to these varying field strengths, the moments in the protons of the sample precess at different rates. These varying precessions mean that the moments get misaligned. The receiver measures the outputted voltage caused by the magnetic moments in the xy plane. As the magnetic moments get out of phase, there becomes an equal number of moments in all directions at all times, due to the number of atoms contained within the sample. Moments in opposite directions destructively interfere, meaning that they produce no signal at all. As the moments of all the atoms get 'shuffled' due to the differing rates of precession, the signal falls back to 0. This is the decay of the envelope signal we see in Figure (2).

We can measure the constant T_2^* using this decay rate. By exporting the oscilloscope data, we can then plot and fit the curve to an exponential. This curve will be defined within the limits of (i) and (ii) as seen in Figure (2), which were determined purely visually and therefore will have a high degree of error. We know that the rate of decay should follow the following exponential:

$$M_{x/y}(t) = M_0 e^{-\frac{t}{T_2^*}} \quad (4)$$

Where the height of the curve is analogous to moment M , and is measured

in time t .

When plotting the data, we will end up with an exponential of the form $y = Ae^{bx}$. We can find the slope constant b of this exponential, then equate it to the slope of the function in equation (4), then simply solve for $T2^*$.

The data was collected 3 separate times, who's slopes gave an average value of $T2^* = (2.8832 \pm 0.0033) * 10^{-5} s$.

This value sets a baseline, as it should be much smaller than and so can be compared to the $T1$ and $T2$ values to be measured.

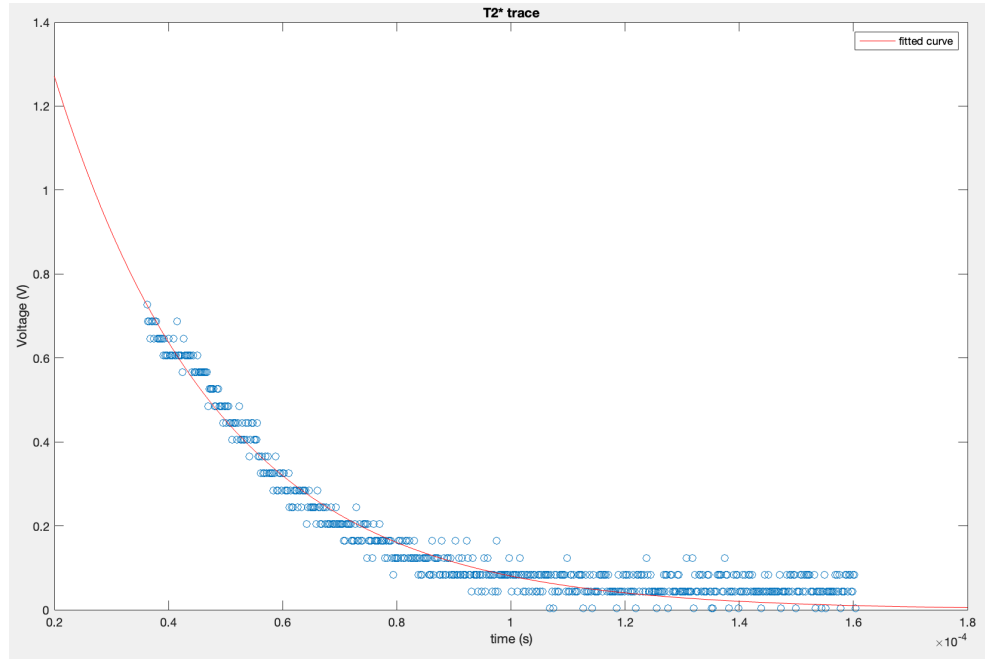


Figure 4: Decay of envelope of signal

3.2 T2: Spin-Spin Relaxation

The spin-spin interaction can be measured by observing the height of spin echoes at different times. This relationship is known to be exponential with time constant T2, so we can fit the data to that kind of curve.

Delay Time (ms)	0.40	0.80	1	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00	20.00
Height (mV) $\pm 20mV$	775	731	718	631	556	506	437	406	356	325	281	256	93

Table 1: spin echo height data as a function of delay time

This data can be fit to the exponential $M_0 e^{-t/T_2}$

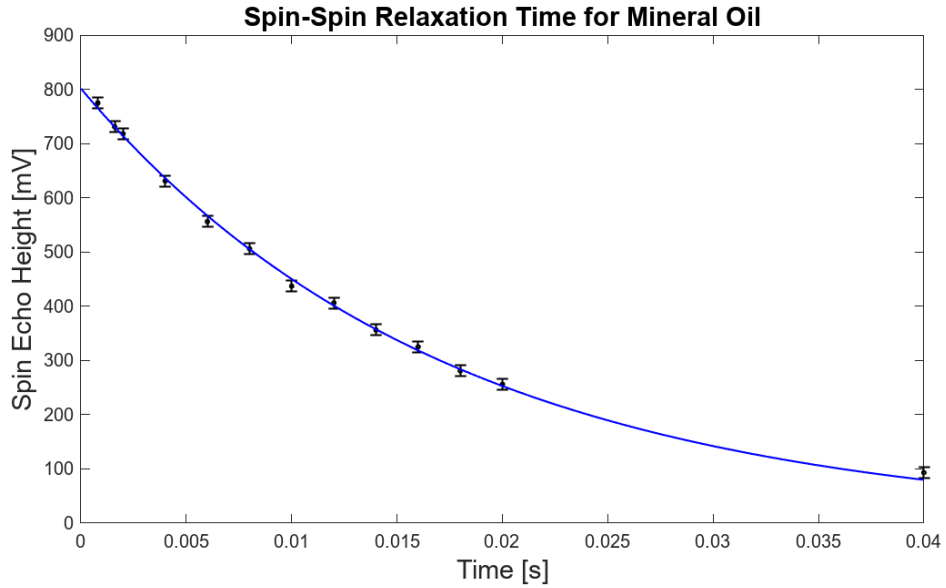


Figure 5: spin echo height as a function of delay time, in order to determine the spin-spin relaxation time

The measured T2 value is 0.017 ± 0.003 s, which makes sense in comparison to the measured value of T2*, as it is much larger.

3.3 T1: Spin-Lattice Relaxation

To measure the spin-lattice relaxation time we perform a 90 degree pulse and a 180 degree pulse to measure the z-component of the total magnetic moment after it has had some time to decay.

Delay (ms)	0.1	1	2	3	4	5	6	7	8	9
Height (mV) $\pm 20mV$	1225	1188	1119	1038	956	900	806	750	656	581
Delay (ms)	10	20	30	40	50	60	70	80	90	99
Height (mV) $\pm 20mV$	492	152	200	500	737	918	1000	1100	1188	1219

Table 2: spin echo height data as a function of delay time

The equation relating these variables is:

$$M_z = M_0(1 - 2e^{-\frac{t}{T_1}})$$

the z-component is inversely exponentially related to time, with time constant T1. The data was fit to the curve, generating the following graph.

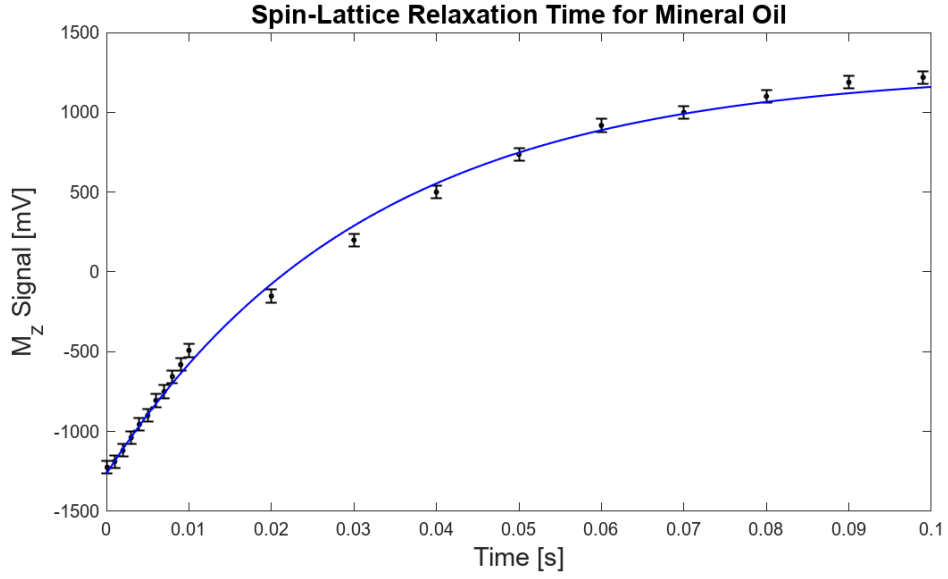


Figure 6: Z-component of the total magnetic moment as a function of time in order to find the spin-lattice relaxation time

The measured T1 value is 0.032 ± 0.004 s, which makes sense in comparison to the measured values of T2* and T2, as it is even longer.

3.4 Summary and Discussion

The measured values of T1, T2, and T2* can be compared to understand better the internal interactions of atoms.

T2*	0.0288 ± 0.0003 ms
T2	0.017 ± 0.003 s
T1	0.032 ± 0.004 s

Table 3: measured relaxation times

The relative sizes of these values provide insight to the strengths of these interactions. The longer the decay, the weaker the interaction, and so the relaxation times easily show the relative strengths of the interactions. The spin-lattice interaction has the longest relaxation time and is therefore the weakest (and consequently the hardest to observe). Next is T2, the spin-spin relaxation time, and finally the strongest effect is the inhomogeneity of the field that causes a very fast evening out of the spins.

This is a quantum example of the experiment in [3], in which a macroscopic magnetic moment is observed in precession. In this case the tiny microscopic nuclear moments precess, much faster and with much smaller magnitude. There is also such a large number of these moments that statistical mechanics comes into play when determining the physics of the system. The macroscopic case provides classical insight into how the spins would interact with magnetic fields which can be useful when analyzing the quantum case.

The applications of pulsed nuclear magnetic resonance are far reaching and allow scientists to gain information on the internal interactions of atoms and molecules, including spin-spin and spin-lattice interactions.

References

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